CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 2: Exponents (Powers)

Topic-wise short notes with solved examples

Powers with integer and simple fractional exponents, the laws of exponents and their use in simplifying expressions and solving for an unknown exponent.

1 Powers

- a^n means a multiplied by itself n times. a is the **base** and n is the **exponent** (index or power).
- $a^1 = a$. **Zero exponent**: $a^0 = 1$ for every $a \neq 0$.
- **Negative exponent**: $a^{-n} = \frac{1}{a^n}$ and $\frac{1}{a^{-n}} = a^n$.
- For a fraction: $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^{n}$.
- A negative base with an **even** exponent gives a positive result; with an **odd** exponent, a negative result.

Example 1.1 Evaluate $(-3)^4$, $(-2)^5$, 7^0 and 4^{-3}.

SOLUTION

81, -32, 1 and $\frac{1}{64}$

Example 1.2 Evaluate $\left(\frac{2}{3}\right)^{-3}$.

SOLUTION

$\left(\frac{3}{2}\right)^{3}$

$\frac{27}{8}$

2 Laws of exponents

- For non-zero bases a, b and integers m, n:

Law	Statement
Product	$a^m \times a^n = a^{m+n}$
Quotient	$a^m \div a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Power of a product	$(ab)^n = a^n \times b^n$
Power of a quotient	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

- The product and quotient laws need the **same base**.
- If $a^m = a^n$ and a is not 0, 1 or -1, then $m = n$.

Example 2.1 Simplify $2^5 \times 2^{-8} \times 2^6$.

SOLUTION

2^{5-8+6}

$2^3 = 8$

Example 2.2 Simplify $\left(\frac{3}{4}\right)^5 \div \left(\frac{3}{4}\right)^3$.

SOLUTION

$\left(\frac{3}{4}\right)^2$

$\frac{9}{16}$

Example 2.3 Simplify $\frac{(5^2)^3 \times 5^4}{5^7}$.

SOLUTION

5^{6+4-7}

$5^3 = 125$

Example 2.4 Simplify $(2^{-1} + 3^{-1})^{-1}$.

SOLUTION

$\left(\frac{1}{2} + \frac{1}{3}\right)^{-1} = \left(\frac{5}{6}\right)^{-1}$

$\frac{6}{5}$

Example 2.5 Simplify $\left(\frac{1}{3}\right)^{-2} - \left(\frac{1}{2}\right)^{-3} + 4^0$.

SOLUTION

$9 - 8 + 1$

2

3 Fractional exponents

- $a^{\frac{1}{2}} = \sqrt{a}$ and $a^{\frac{1}{3}} = \sqrt[3]{a}$. In general $a^{\frac{1}{n}} = \sqrt[n]{a}$.
- $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$: take the root first, then the power.
- The laws of exponents hold for fractional exponents too.

Example 3.1 Evaluate $(64)^{\frac{2}{3}}$ and $(81)^{\frac{3}{4}}$.

SOLUTION

$(4^3)^{\frac{2}{3}} = 4^2$ and $(3^4)^{\frac{3}{4}} = 3^3$

16 and 27

Example 3.2 Evaluate $\left(\frac{27}{125}\right)^{-\frac{2}{3}}$.

SOLUTION

$\left(\frac{125}{27}\right)^{\frac{2}{3}} = \left(\frac{5}{3}\right)^2$

$\frac{25}{9}$

4 Expressing in exponential form

- Write the number as a product of prime factors and collect equal factors.

Example 4.1 Express 432 as a product of powers of primes.

SOLUTION

$2^4 \times 3^3$

Example 4.2 Simplify $\frac{16 \times 2^{n+1} - 4 \times 2^n}{16 \times 2^{n+2} - 2 \times 2^{n+2}}$.

SOLUTION

Numerator: $2^n(32 - 4) = 28 \times 2^n$. Denominator: $2^n(64 - 8) = 56 \times 2^n$.

$\frac{1}{2}$

5 Finding an unknown exponent

- Make the bases equal on both sides, then equate the exponents.

Example 5.1 Find x if $3^{x+1} = 81$.

SOLUTION

$3^{x+1} = 3^4$

$x = 3$

Example 5.2 Find x if $\left(\frac{2}{5}\right)^{3} \times \left(\frac{2}{5}\right)^{-6} = \left(\frac{2}{5}\right)^{2x-1}$.

SOLUTION

$3 - 6 = 2x - 1$

$x = -1$

Example 5.3 Find x if $4^x = 8^{x-1}$.

SOLUTION

$2^{2x} = 2^{3x-3}$, so $2x = 3x - 3$

$x = 3$

Example 5.4 By what number should $(-8)^{-1}$ be multiplied to get 10^{-1}?

SOLUTION

Required number $= \frac{1}{10} \div \frac{-1}{8} = \frac{1}{10} \times (-8)$

$-\frac{4}{5}$

6 Standard form

- A number in **standard form** is written as $k \times 10^n$, where $1 \leq k < 10$ and n is an integer.
- Large numbers have positive n; small numbers (less than 1) have negative n.

Example 6.1 Write 3,84,000 and 0.00056 in standard form.

SOLUTION

3.84×10^5 **and** 5.6×10^{-4}

Example 6.2 Write 2.07×10^{-3} in the usual form.

SOLUTION

0.00207

7 Common mistakes

- Writing $2^3 = 6$. It is $2 \times 2 \times 2 = 8$.
- Taking a^{-n} as a negative number.
- Writing $a^0 = 0$.
- Adding exponents with different bases, or multiplying exponents where they should be added.
- Writing $(a + b)^n = a^n + b^n$.
- Confusing $(-2)^4 = 16$ with $-2^4 = -16$.

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